

KOTA KINABALU INTL, Malaysia

WMO: 964710

Lat: 5.937N

Lon: 116.051E

Elev: 3

StdP: 101.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	23.0	19.9	14.6	24.9	20.6	15.3	25.1	5.9	27.7	5.3	27.8	2.4	100	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.9	34.1	28.3	33.7	28.1	33.1	27.7	29.0	33.3	28.6	32.9	28.2	32.5	2.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.0	32.9	27.2	23.0	32.2	27.0	22.8	32.1	94.9	33.1	92.6	33.3	91.3	32.4	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.0	4.3	21.3	35.2	1.2	0.9	20.5	35.8	19.8	36.3	19.2	36.8	18.3	37.4		
(5)			19.6	30.0	1.1	0.9	18.7	30.6	18.1	31.1	17.4	31.6	16.6	32.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	27.4	27.4	27.9	28.6	28.8	28.5	28.2	28.3	28.1	27.9	27.9	27.8
	DBStd	0.99	0.90	0.86	0.85	0.79	0.96	0.97	0.93	0.97	0.86	0.91	0.77	0.87
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6595	538	487	556	559	584	554	564	566	543	555	537	551
	CDD18.3	3553	280	253	298	309	326	304	306	308	293	297	287	293
	CDH23.3	36968	2653	2452	3097	3459	3675	3287	3255	3328	3072	2959	2848	2882
	CDH26.7	14123	874	816	1146	1427	1547	1323	1306	1348	1196	1091	1029	1020
Wind	WSAvg	2.1	1.9	1.8	1.8	2.0	2.1	2.1	2.2	2.3	2.2	2.2	2.0	2.0
Precipitation	PrecAvg	2621	136	73	81	118	217	281	262	263	290	357	288	237
	PrecMax	3975	703	277	409	323	492	671	638	623	659	678	638	620
	PrecMin	1795	0	0	0	9	29	57	60	67	95	31	63	1
	PrecStd	510	133	72	82	85	104	144	122	126	107	142	120	164
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	32.8	33.2	34.2	34.9	34.6	34.2	34.2	33.9	33.7	33.2	33.1
		MCWB	27.5	27.2	27.4	28.5	28.9	28.4	28.2	27.8	28.1	28.0	28.2	27.9
	2%	DB	32.0	32.0	32.6	33.2	34.0	33.8	33.3	33.2	33.1	32.9	32.6	32.2
		MCWB	26.9	26.8	27.2	28.0	28.4	28.0	27.7	27.6	27.7	27.8	27.7	27.3
	5%	DB	31.2	31.2	32.0	32.9	33.1	32.9	32.7	32.8	32.2	32.1	32.0	31.9
		MCWB	26.6	26.5	26.9	27.8	28.0	27.4	27.3	27.4	27.1	27.2	27.3	27.0
	10%	DB	30.6	30.7	31.1	32.1	32.4	32.0	31.9	32.0	31.8	31.2	31.2	31.1
		MCWB	26.3	26.3	26.6	27.3	27.5	26.9	26.8	26.9	26.9	26.8	26.9	26.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.1	28.5	29.3	29.6	29.1	29.0	29.2	28.9	29.0	28.9	28.6
		MCDB	32.0	31.9	32.4	33.2	33.9	33.8	33.5	33.5	33.2	32.9	32.8	32.5
	2%	WB	27.4	27.3	27.7	28.7	28.8	28.4	28.2	28.3	28.2	28.2	28.1	27.9
		MCDB	31.3	31.1	31.8	32.8	33.3	32.9	32.7	32.6	32.4	32.3	32.1	31.8
	5%	WB	26.8	26.8	27.2	28.1	28.3	27.8	27.6	27.7	27.6	27.5	27.5	27.3
		MCDB	30.6	30.5	31.2	32.3	32.7	32.1	32.0	31.9	31.8	31.6	31.5	31.2
	10%	WB	26.3	26.3	26.8	27.6	27.7	27.2	27.1	27.1	27.0	26.9	27.0	26.7
		MCDB	29.8	29.9	30.6	31.7	32.0	31.3	31.2	31.2	31.1	30.8	30.9	30.4
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.1	7.5	7.9	7.9	7.8	7.9	7.8	7.8	7.5	7.5	7.3
		MCDBR	7.8	7.8	8.0	8.5	8.6	8.7	8.8	8.7	8.4	8.3	8.2	8.1
		MCWBR	4.2	4.3	4.5	5.0	5.1	5.1	5.4	5.4	5.1	4.9	4.8	4.5
	5% WB	MCDBR	7.5	7.5	7.7	8.2	8.4	8.3	8.4	8.3	8.2	8.1	8.0	7.8
		MCWBR	4.2	4.3	4.5	5.0	5.1	5.0	5.4	5.3	5.2	4.9	4.8	4.5
Clear-Sky Solar Irradiance	taub	0.407	0.414	0.419	0.428	0.419	0.421	0.417	0.450	0.451	0.457	0.418	0.408	
	taud	2.510	2.466	2.451	2.421	2.463	2.476	2.481	2.364	2.357	2.345	2.470	2.510	
	Ebn at Noon	912	912	905	881	870	858	865	851	863	861	894	904	
	Edh at Noon	109	116	118	120	112	109	109	125	128	129	112	107	
All-Sky Solar Radiation	RadAvg	4.61	5.33	5.76	5.98	5.53	5.15	5.12	5.28	5.16	4.87	4.79	4.50	
	RadStd	0.46	0.52	0.47	0.32	0.29	0.27	0.25	0.28	0.34	0.34	0.31	0.42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.20	N/A	N/A	+0.70	+0.57	+0.54	N/A	N/A	+114
(47) Regional (3 neighbors)		+0.13	N/A	N/A	+0.28	+0.30	+0.25	N/A	N/A	+82

Nomenclature: See separate page